

Course Code: CSE-2421
Course Title: Computer Algorithms
Credit Hours: 3
Contact Hours: 3 lecture hours per week
Type: Core, Engineering
Prerequisite: CSE-2321 (Data structures)
Co-requisite: CSE-2422 (Computer Algorithms Lab)
Instructor's Name and Office Location

Instructor: Mysha Sarin Kabisha

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Course Assessments	CIE: Continuous Internal Evaluation	Attendance	10 Marks
		Class test/ Assignment/ Quizzes	10 Marks
		Mid-term	30Marks
	SEE: Semester End Examination		50 Marks

This course introduces students to the general tools and techniques for analyzing and designing computer algorithms. Initially necessary mathematical preliminaries required for analyzing and designing computer algorithms are taught. Then this course familiarizes students with several algorithmic approaches and corresponding problems. To take this course, students should be able to program in a standard programming language preferably in C/C++. They should have familiarity with basic data structures. Some mathematical maturity also is expected; students should have some idea of what constitutes a mathematical proof and how to write one. Some knowledge of basic probability will also be helpful. Knowledge of this course will be needed as prerequisite knowledge for future courses such as Artificial Intelligence, Computer Networks and Compiler.

Course Learning Outcomes (CLOs):

Upon successful completion of this course, students will be able to:

#	CLO Description	Weightage (%)
1.	Demonstrate a familiarity with major algorithms and data structures	10%
2.	Analyze algorithms for understanding and comparing their working mechanism	20%
3.	Analyze algorithms for deriving proof of correctness	40%
4.	Analyze the asymptotic performance of algorithms	20%
5.	Analyze and apply important algorithmic design techniques and methods of analysis	10%

Mapping of CLO-PLO:

#	CLO Description	PLOs	Bloom's Taxonomy Domain/Level	Delivery Methods and activities	Assessment Tools

#	CLO Description	PLOs	Bloom's Taxonomy Domain/Level	Delivery Methods and activities	Assessment Tools
CLO1	Demonstrate a familiarity with major algorithms and data structures	PLO1	Cognitive/Understand	Lecture, Class discussion, Assignment,	Exam, Quiz, Assignment
CLO2	Analyze algorithms for understanding and comparing their working mechanism	PLO1	Cognitive/Analyze	Lecture, Class discussion, Assignment, Note	Exam, Quiz, Assignment
CLO3	Analyze algorithms for deriving proof of correctness	PLO2	Cognitive/Analyze	Lecture, Class discussion, Assignment, Note	Exam, Quiz, Assignment
CLO4	Analyze the asymptotic performance of algorithms	PLO2	Cognitive/Analyze	Lecture, Class discussion, Assignment, Note	Exam, Quiz, Assignment
CLO5	Analyze and apply important algorithmic design techniques and methods of analysis	PLO1 PLO2	Cognitive/Analyze	Lecture, Class discussion, Assignment	Exam, Quiz, Assignment,

Course Content:

Section-A (Mid-term: 30 Marks)

1. Algorithm and Data structure: Introduction to algorithm; Properties of good algorithm; Correctness proof and techniques for time and space complexity analysis of algorithms (using insertion-sort as an example); Application areas of algorithm; Growth of functions and asymptotic notations

2. Divide and Conquer approach Sorting and Heaps: Divide and Conquer approach; Asymptotic bound of recursive algorithm by solving recurrence, substitution method, recursion tree method, master method; Maximum subarray problem; Merge-sort and complexity analysis; Quick-sort, Randomized Quick-sort and complexity analysis; Heap, Heap construction algorithm, Heap-sort, Priority queue, complexity analysis of related algorithms; Sorting in Linear Time, Lower bound of sorting, Counting sort, Radix sort.

3. Dynamic Programming: Dynamic programming, Elements of dynamic programming, Memoization, Matrix-chain multiplication and longest common subsequence problems as examples, Complexity analysis of the algorithms.

Section-B (Final Exam: 50 Marks)

Group-A (20 Marks)

4. Greedy Algorithms and String Matching Algorithms: Greedy algorithms, Activity selection problem, Elements of greedy strategy, Huffman codes and its application; String Matching Algorithms, Naive string-matching algorithm, Rabin-Karp algorithm; Complexity analysis of the algorithms

5. Graphs Basic & Traversal Techniques: Representation of Graphs, Breadth First Search, Depth First Search, Algorithm of BFS and DFS, Application of BFS and DFS, Minimum Spanning Tree, Kruskal's and Prim's Algorithm, Complexity analysis of the algorithms

Group-B (30 Marks)

6. Shortest Path Algorithms: Single-source shortest path, Dijkstra's Algorithm, Bellman-Ford's Algorithm; All-pairs shortest path, Floyd-Warshall's Algorithm; Complexity analysis of the algorithms

7. Computational Geometry & Number Theory: Computational Geometry, Line Segment Properties, Convex Hull, Graham Scan Algorithm of Convex Hull, Number Theory, GCD, Modular Arithmetic, Prime Number generation, Complexity analysis of the algorithms

8. Theory of NP-Completeness and Coping with Hardness: Theory of NP-Completeness, P, NP, NP-Complete and NP-Hard Problems; Backtracking, N-Queen Problem; Branch and Bound; Approximation algorithms

Course Assessment Pattern:

Bloom's taxonomy domain/level	Evaluation out of 100 marks				
	CIE (50 marks)			SEE (50 marks)	Total
	Attendance (10)	Assignment/Class Test (10)	Midterm (30)	Written Exam (50)	Total (100)
Cognitive/Remember	-	-	-	-	-
Cognitive/Understand	-	2	5	5	12
Cognitive/Apply	-	5	10	20	35
Cognitive/Analyze	-	3	5	15	23
Cognitive/Evaluation	-	-	5	5	10
Cognitive/Create	-	-	5	5	10
Affective/Responding	10	-	-	-	10

Text Books:

1. Thomas H. Cormen, Charles E. Leiserson, Ronald L. Rivest and Clifford Stein, Introduction to Algorithms, 3rd Edition, MIT Press, 2009, ISBN-13: 978-0262033848
2. Ellis Horowitz, Sartaj Sahni and Sanguthevar Rajasekaran, Fundamentals of Computer Algorithms, 1st Edition, Orient Black Swan, 2008, ISBN-13: 978-8173716126

Reference Books:

3. Robert Sedgewick, Algorithms, 4th Edition, Addison-Wesley Professional, 2011, ISBN-13: 978-0321573513
4. Steve Skiena, The Algorithm Design Manual, 2nd Edition, Springer, 2008, ISBN-13: 978-1849967204
5. Robert L. Kruse, Alexander J. Ryba, Data Structures and Program Design in C++, 1st Edition, Prentice Hall, 1998, ISBN-13: 978-0137689958
6. Mark Allen Weiss, Data Structure and Algorithm analysis in C++, 4th Edition, Pearson, 2013, ISBN-13: 978-0132847377
7. Jeff Edmonds, How to Think About Algorithms, 1st Edition, Cambridge University Press, 2014, ISBN-13: 978-1107439887
8. Steven Halim and Felix Halim, Competitive Programming 3, 3rd Edition, Lulu.com, 2013
9. Donald Ervin Knuth, The Art of Computer Programming, Volume 1, Addison-Wesley Professional, 3rd edition, 1997, ISBN-13: 978-0201896831
10. Donald Ervin Knuth, The Art of Computer Programming, Volumes 1, 2, 3, 4A, Addison-Wesley Professional, 1st edition, 2011, ISBN-13: 978-0321751041

Online Resources:

I) <https://www.topcoder.com/community/competitive-programming/tutorials/>

Grading Policy:

As per IIUC grading policy

Weekly Activity Plan:

Week	Activities	Topics
1	Lecture (3)	Introduction to algorithm; Properties of good algorithm; Correctness proof and techniques for time and space complexity analysis of algorithms (using insertion-sort as an example); Application areas of algorithm;
2	Lecture (3)	Growth of functions and asymptotic notations; Divide and Conquer approach; Merge-sort and complexity analysis;
3	Lecture (3) Quiz 1	Asymptotic bound of recursive algorithm by solving recurrence, substitution method, recursion tree method, master method; Review
4	Lecture (3)	Quick-sort, Randomized Quick-sort and complexity analysis; Heap, Heap construction algorithm, Heap-sort, Priority queue, complexity analysis of related algorithms;
5	Lecture (3)	Sorting in Linear Time, Lower bound of sorting, Counting sort, Radix sort; Dynamic programming, Matrix-chain multiplication;
6	Lecture (3) Quiz 2	Matrix-chain multiplication and longest common subsequence, Elements of dynamic programming, Memoization; Review
7	Lecture (3)	Greedy algorithms, Activity selection problem, Elements of greedy strategy, Huffman codes and its application;
8	Lecture (3)	String Matching Algorithms, Naive string-matching algorithm, Rabin-Karp algorithm;
9	Lecture (3)	Representation of Graphs, Breadth First Search, Depth First Search, Algorithm of BFS and DFS, Application of BFS and DFS
10	Lecture (3)	Minimum Spanning Tree, Kruskal's and Prim's Algorithm Review
11	Lecture (3) Quiz 3	Single-source shortest path, Dijkstra's Algorithm, Bellman-Ford's Algorithm;
12	Lecture (3)	All-pairs shortest path, Floyd-Warshall's Algorithm; Complexity analysis of the algorithms Review
13	Lecture (3)	Computational Geometry, Line Segment Properties, Convex Hull, Graham Scan Algorithm of Convex Hull
14	Lecture (3)	Number Theory, GCD, Modular Arithmetic, Prime Number generation, RSA Review
15	Lecture (3) Quiz 4	Theory of NP-Completeness, P, NP, NP-Complete and NP-Hard Problems; Backtracking, N-Queen Problem; Branch and Bound; Approximation algorithms